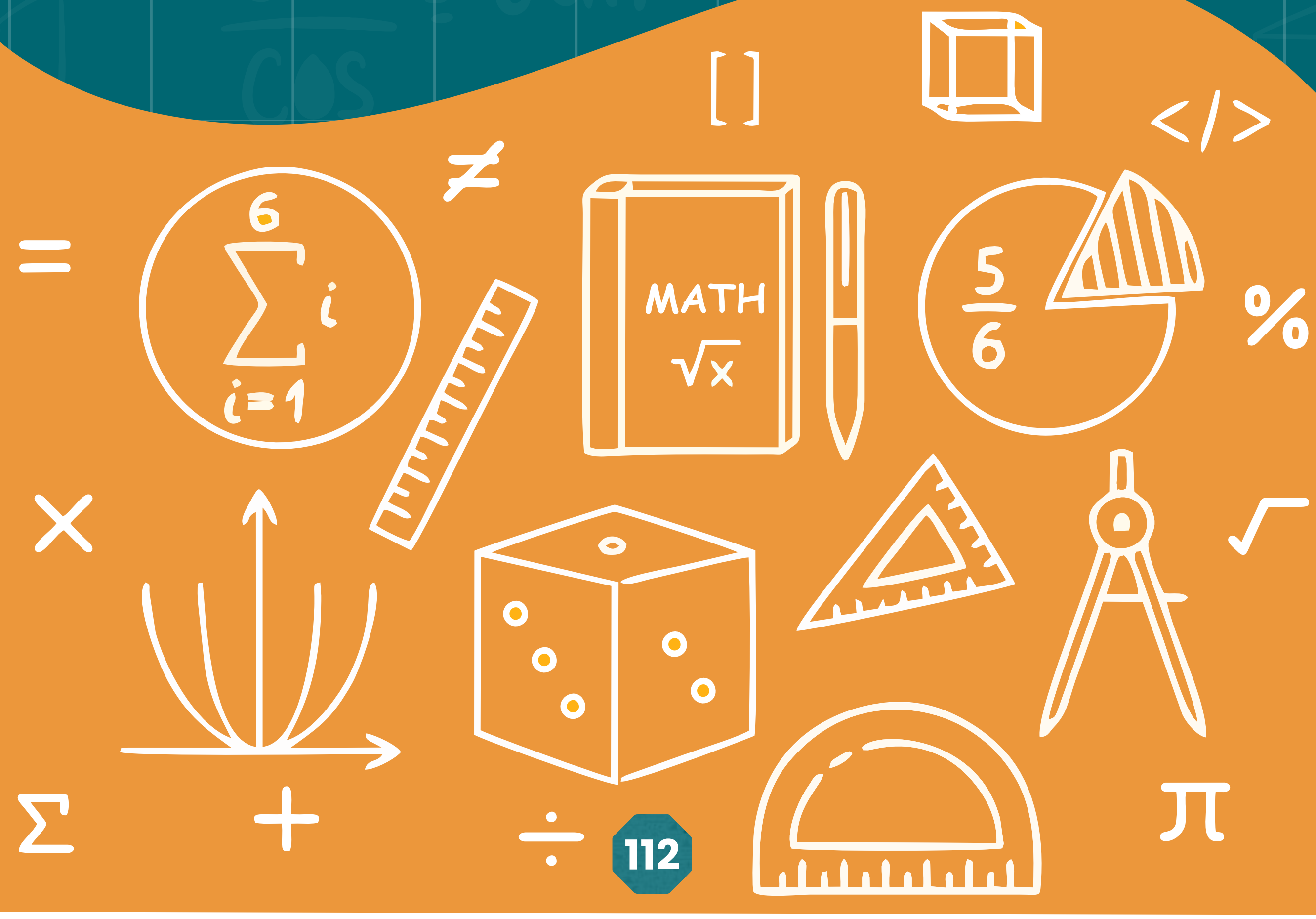
UNIT 10

DEMONSTRATIVE GEOMETRY



'Demonstrative Geometry' is a branch of Mathematics in which statements concerning geometrical figures (called theorems) are proved through logical reasoning.
The word 'Demonstrate' comes from the Latin word **'Demonstratum'** which means **'to prove with certainty'**.

Describe the Basics of Reasoning
Reasoning is the process of thinking about something in a logical way in order to form a conclusion or judgment. Reasoning may be Inductive or Deductive.

Inductive Reasoning:
If we see a large number of school buses in a country that have yellow colour and do not see any bus of different colour, we conclude that all school buses in that country are of yellow colour. Such a reasoning, in which we generalize specific observations into universal statements, is called Inductive Reasoning.

Deductive Reasoning:
Contrary to this, in Deductive Reasoning we deduce particular results from general ones. Generally we start from a set of statements already believed to be true and construct a series of deductions and use to establish proof for new statements.
We study Geometry through Deductive Reasoning because the new statements can be deduced with certainty.
The whole framework of deductive reasoning is based on the following four elements known as the Basics of Reasoning.

1. Undefined Terms: Some geometrical terms are accepted without definition. These are called Undefined Terms.
Point, Line, Plane, and Space are Undefined Terms.
2. Defined Terms: With the help of undefined terms, we define various new geometrical terms. **For example:** Line Segment, Ray, Angle, Triangle, Perpendicular, Bisector etc. are defined terms. Many of these terms have been defined in earlier classes.

AXIOMS, POSTULATES AND THEOREM

Describe the Types of Assumptions (Axioms and Postulates):
Assumptions or Fundamental Agreements are of two kinds:
Axioms: Fundamental Agreements which are related to numbers are called Axioms. Following are the axioms which will be used in this unit later on.
Axiom 1: Every number is equal to itself (Reflexive Property of Equations)
i.e., $x = x \ \forall x \in \mathbb{R}$
or $\angle A = \angle A$ (Identity Congruence)
Axiom 2: Symmetric Property of Equations, i.e.,

$x = y \Leftrightarrow y = x \ \forall \ x, y \in \mathbb{R}$

Axiom 3: Transitive Property of Equations, i.e.,
$x = y$ and $y = z \Rightarrow x = z \ \forall \ x, y, z \in \mathbb{R}$

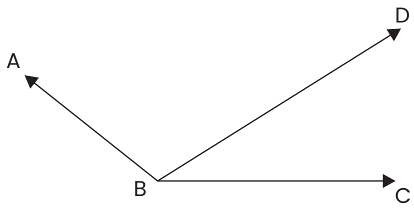
Axiom 4: Equals added to equals remain equal (Addition Property of Equations) i.e.,
$x = y \Rightarrow x + z = y + z \ \forall \ x, y, z \in \mathbb{R}$

Axiom 5: Equals subtracted from equals remain equal (Subtraction Property of Equations) i.e.,
$x = y \Rightarrow x - z = y - z \ \forall \ x, y, z \in \mathbb{R}$

Axiom 6: If equals are multiplied by equals, their products are equal (Multiplication Property of Equations) i.e.,
$x = y \Rightarrow xz = yz \ \forall \ x, y \in \mathbb{R}$

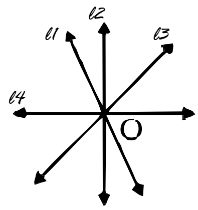
Axiom 7: If equals are divided by a non-zero number, their quotients are equal i.e.,
$x = y$ and $z \neq 0 \Rightarrow x / z = y / z \ \forall \ x, y \in \mathbb{R}$

Axiom 8: Whole is greater than its part. i.e.,
$m\angle ABC = m\angle ABD + m\angle CBD$
$\Rightarrow m\angle ABC > m\angle ABD, m\angle ABC > m\angle CBD$



B. Postulates: Fundamental Agreements which are related to geometrical figures are called Postulates.

Postulate 1. One and only one line passes through two distinct points.
It is also stated as: "Two points determine a line."



Postulate 2. Countless lines can pass through a point.
Lines $l, l_1, l_2, \ldots, l_n$ are passing through a point O.

Postulate 3. One and only one plane passes through three points not in a line.

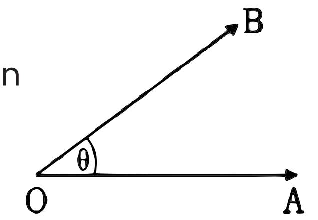
Postulate 4. If two points of a line lie on a plane, the whole line lies on the plane.
[This postulate tells us that the surface of the plane is smooth and it is infinite on all sides.]

Postulate 5 (Distance Postulate). If A and B are any two points of a plane, then the distance between them will be:
(i) 0 (zero), if A lies on B, i.e., A = B [read as A coincides with B]
(ii) a Positive real number when A and B are distinct points.
The distance from point A to point B is denoted as $|AB|$ or mAB,
m stands for 'measure.' It may be noted that $mAB = mBA$ or $|AB| = |BA|$.

Postulate 6. A line segment can be bisected at one and only one point.

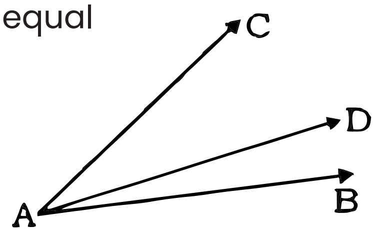
[This postulate tells that on a line segment AB, there is only one point (say P) between A and B such that mAP = mBP]

Postulate 7 (Angle Construction Postulate).



If one arm of an angle is given, one and only one ray can be drawn on one side of it making an angle of given measure θ between 0° and 180°. i.e., $0° < \theta < 180°$.

Postulate 8 (Angle Addition Postulate).

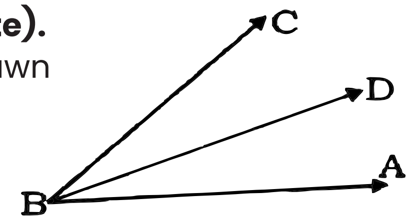


The sum of the measures of two adjacent angles is equal to the measure of the angle formed by their non-common arms.

In this figure, ∠BAD and ∠CAD are two adjacent angles. So,

m∠BAD + m∠CAD = m∠BAC (or m∠CAB).

Postulate 9 (Unique Bisector of an Angle Postulate).



One and only one bisector of an angle can be drawn

In this figure BD is the bisector of ∠ABC

So

m ∠ABD = m ∠CBD = ½ m ∠ABC

Postulate 10

At a given point on a line in a plane, one and only one perpendicular can be drawn to the line.

Postulate 11

Geometrical figures which can be made to coincide with one another are congruent.

Postulate 12

A geometrical figure may be moved from one position to another without any change in shape and size.

Postulate 13 (Playfair's Axiom)

Through a point not on a line, one and only one line can be drawn parallel to the line on a particular plane. It is also stated as 'Two intersecting lines cannot be parallel to a third line'.

Describe parts of a proposition

When a statement about any geometrical subject is proposed for discussion, it is called a **proposition**.

Propositions are either **Theorems or Problems** (also called Riders).

A Proposition (Theorem or **Rider**) consists of two parts:

(i) **Hypothesis** or that which needs to be checked if it is true.

(ii) **Conclusion** which shows hypothesis is true or not.

Meanings of geometrical terms:

(a) Theorem

A Theorem is a proposition in which some geometrical fact is stated. The general statement of a theorem is called Enunciation e.g.

'If two lines intersect then the opposite vertical angles are congruent'

(b) Corollary

A Corollary (abbreviated as Cor) is a fact which readily follows from a proved theorem. Such propositions are so trivial that we do not label them as separate theorems. These are the deductions made from a proposition or a theorem.

(c) Converse of a Theorem

One theorem is said to be the Converse of the other when the hypothesis of each is the conclusion of the other.

In order to prove a Theorem, a Corollary or a Rider {given as an exercise) we follow certain fixed steps which are as under:

1. Figure or Diagram: According to the Enunciation, a neat figure is drawn showing the requirements of the Hypothesis as well as that of the Conclusion.

2. Given or Data: This is the particular enunciation of the hypothesis and is interpreted in terms of the figure in Step 1 above. It tells us what is the hypothesis in particular.

3. Required to Prove: Here the Conclusion is written in particular terms according to the figure drawn in Step 1 above.

4. Construction: Sometimes we need certain addition in the figure that helps us in proving the theorem. So we do certain construction.

5. Proof: This is the last part of a proof of a theorem {or a Problem) in which the given proposition is proved logically with the help of definitions, postulates, axioms, and the given data, or theorems proved earlier.

6. At the end of a Proof, we usually write Q.E.D., the abbreviation of Quod Erat Demonstrandum meaning 'which was to be proved'.

EXERCISE 10.1

1. Fill in the blanks with appropriate words:

(a) Demonstrative Geometry is a ___________ of Mathematics in which ___________ on geometry are proved through ___________.

(b) Reasoning may be ___________ or ___________ .

(c) In ___________ reasoning we draw conclusions based on observations.

(d) In ___________ reasoning we deduce particular results from general results and use ___________ to prove new statements.

(e) Assumptions are those statements which are accepted ___________.

(f) Fundamental Agreements are of two kinds 1 ___________ 2 ___________.

(g) Fundamental Agreements which are related to geometrical figures are called ___________.

(h) The four basics of Reasoning are:

1 ___________ 2 ___________ 3 ___________ 4 ___________

2. Write 'True' or 'False' against each statement:

(1) One and only one straight line can be drawn through three distinct points. (   )

(2) If P and Q are coincident points, symbolically we write P = Q.(   )

(3) Two points do not determine a line. (   )

(4) Postulate 4 tells us that the surface of the plane is smooth and it is infinite on all sides. (   )

(5) Complement of an angle of x^0 is $(90 - x)^{\circ}$. (   )

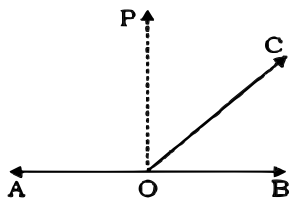
(6) 'If two quantities are equal to the same quantity, they are equal to one another' is the same as Transitive Property of Equations. (   )

THEOREMS

Prove the following theorems along with corollaries and apply them to solve appropriate problems.

THEOREM 1

If a straight line stands on another straight line the sum of measures of two angles so formed is equal to two right angles



Given: OC stands on AB

To prove: m ∠AOC + m ∠BOC = 2rt∠s (or 180°)

Construction: Through O Draw OP ⊥ AB

i.e., m∠AOP = 1 rt∠ = m ∠BOP

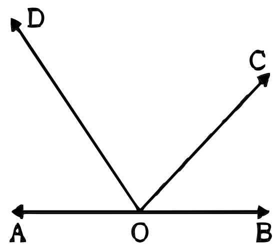
Proof: Statements	Reasons
1. m∠AOC = m∠AOP + m∠POC	i. Angle Addition Postulate
2. m∠BOC = m∠BOP - m∠POC	ii. m∠BOC + m∠POC = m∠POB (AAP)
3. m∠AOC + m∠BOC = m∠AOP + m∠BOP	iii. Adding (1) and (2), m∠POC cancels
= 1 rt∠s + 1 rt∠s	Each is a right angle
= 2 rt∠s (or 180°)	

Q.E.D.

Cor. 1. The sum of measures of all the angles at a point in a given straight line on the same side of it is equal to two right angles

Given: OC and OD stand on AB on the same side of it making angles ∠AOD, ∠DOC and ∠BOC

To Prove: m∠AOD + m∠DOC + m∠BOC = 180°

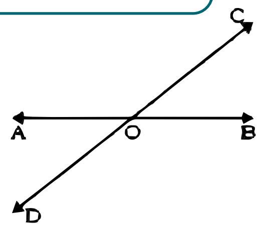


Note: Statements No 1, 2, 3 etc have the reasons i, ii, iii, etc respectively in each and every theorem

Proof: Statements	Reasons
1. m∠AOC + m∠BOC = 180°	i. By Theorem 1
2. But m∠AOC = m∠AOD + m∠DOC	ii. Angle Addition Postulate
3. m∠AOD + m∠DOC + m∠BOC = 180°	iii. Putting the value of m∠AOC in (1)

Q.E.D.

Cor. 2. If two lines intersect, the sum of the measures of the four angles is equal to four right angles



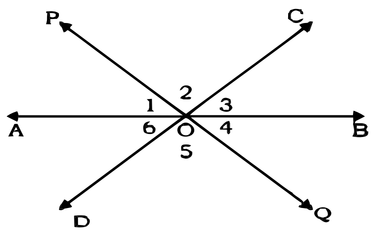
Given: AB and CD intersect at O

To Prove: m∠AOC + m∠BOC + m∠BOD + m∠AOD = 360°

Proof: Statements	Reasons
1. m∠AOC + m∠BOC = 180°	i. By Theorem 1
2. m∠BOD + m∠AOD = 180°	ii. By Theorem 1
3. m∠AOC + m∠BOC + m∠BOD + m∠AOD = 360°	iii. Adding (1) and (2)

Q.E.D.

Cor. 3. If a number of straight lines meet at a point the sum of measures of all the angles between successive lines is equal to four right angles



Given: AB, CD and PQ intersect at point O

To Prove: m∠1 + m∠2 + m∠3 + m∠4 + m∠5 + m∠6
= 360°
= 4 rt∠s

Proof: Statements	Reasons
1. m∠1 + m∠2 + m∠3 = 180°	i. By Theorem 1 Cor. 1
2. m∠4 + m∠5 + m∠6 = 180°	ii. By Theorem 1 Cor. 1
3. m∠1 + m∠2 + m∠3 + m∠4 + m∠5 + m∠6 = 360° = 4 rt∠s	iii. Adding (1) and (2)

Q.E.D.

THEOREM 2

If the sum of measures of two adjacent angles is equal to two right angles the external arms of the angles are in a straight line

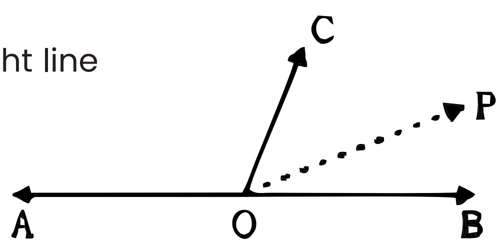
Given:

Two adjacent angles ∠AOC and ∠BOC such that m∠AOC + m∠BOC = 180° = 2 rt∠s

To prove: $\overline{OA}$ and $\overline{OB}$ are in the same straight line

Construction:

If OB is not in the same straight line with OA produce AO to point P



Proof: Statements	Reasons
1. AOP is a straight line and OC stands on it	i. Construction, Supposition
2. $m\angle AOC + m\angle POC = 180^\circ$	ii. By Theorem 1
3. But $m\angle AOC + m\angle BOC = 180^\circ$	iii. Given
4. $m\angle AOC + m\angle POC = m\angle AOC + m\angle BOC$	iv. Axiom 3
5. Or $m\angle POC = m\angle BOC$	v. Canceling $m\angle AOC$ from both sides
6. Which is impossible unless OP and OB coincide	vi. Our Supposition in the construction is wrong. So, given facts are always true.
Hence OA, OB are in the same straight line.	

Q.E.D.

EXERCISE 10.2

1. In the figure of Theorem 1.

(i) Name the angle which is supplement of BOC

(ii) Name the angle which is complement of BOC

(iii) If $m\angle BOC = 50^\circ$, find measures of its complement and supplement.

2. In the figure of Theorem 1 Cor. 2.

(a) Let $m\angle BOC = x = 40^\circ$, find the remaining angles

[**Hint:** Let $m\angle AOC = y$, $m\angle AOD = z$, $m\angle BOD = w$,

Now solve $w + 40^\circ = 180^\circ$, $y + 40^\circ = 180^\circ$, then $z + 140^\circ = 180^\circ$ (why?)]

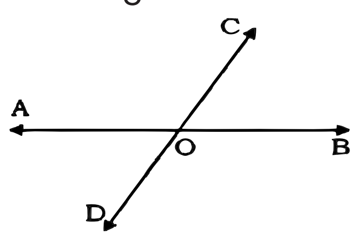
(b) Let $m\angle BOC = x = 45^\circ$, find the remaining angles.

THEOREM 3

If two lines intersect each other then the opposite vertical angles are congruent

Given:

Two lines AB and CD intersect at point O



To Prove:

(i) $\angle BOC \cong \angle AOD$

(ii) $\angle AOC \cong \angle BOD$

Proof: Statements	Reasons
1. $m\angle AOC + m\angle BOC = 180^\circ$	i. By Theorem 1
2. $m\angle AOD + m\angle AOC = 180^\circ$	ii. By Theorem 1
3. $\therefore m\angle AOC + m\angle BOC = m\angle AOD + m\angle AOC$	iii. Transitive Prop. of Eq.
4. $\therefore m\angle BOC = m\angle AOD$	iv. Cancelling $m\angle AOC$ from both sides
5. or $\angle BOC \cong \angle AOD$	v. If two angles are equal in measure, they are congruent.
6. Similarly, we can prove that $\angle AOC \cong \angle BOD$	vi. By the above process.

Q.E.D.

EXERCISE 10.3

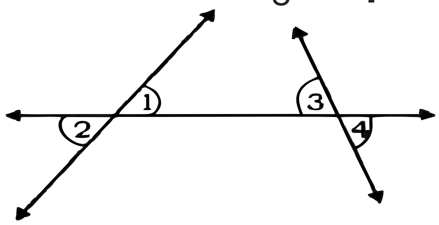
Q1. In the figure of Theorem 3, if m BOC = 70º, find the measures of other angles.

[**Comment:** In the previous exercise we used Theorem 1 to find other angles, because until then we had not learnt Theorem 3. But now we can use Theorem 3 together with Theorem 1]

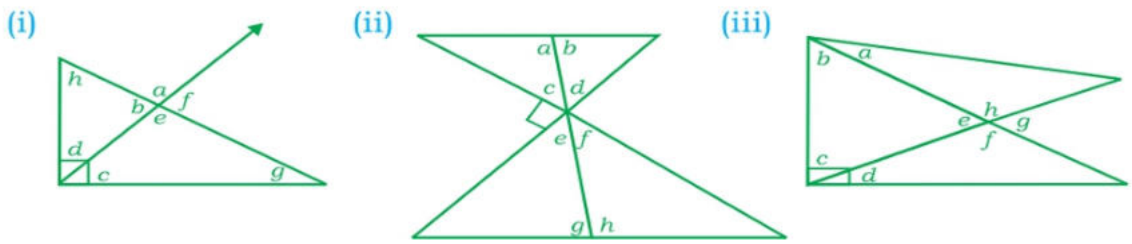
Solution: m∠AOD = m∠BOC = 70º (By Theorem 3) and ∠AOC is the supplement of ∠BOC. Therefore, m∠AOC = 180º - 70º = 110º (By Theorem 1) and m∠BOD = 110º (By Theorem 3).

Q2. Draw two lines intersecting one another at an angle of 30º, find the measures remaining angles as discussed above.

Q3. In this figure ∠1 ≅∠3, Prove that ∠2 ≅∠4 [Hint: First prove ∠2 ≅∠1 and ∠3 ≅∠4 (Th. 1) then use Axioms 3 since ∠1 ≅∠3 is given.]



Q4. In the following figures, name the complementary angles, the supplementary angles, and the vertically opposite angles:



Q5. Find the degree measure for the angles marked x, y and z